

SAULT STE MARIE SW PIER, MI, USA

WMO: 997268

Lat: 46.501N

Lon: 84.372W

Elev: 186

StdP: 99.11

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.9	(c) -20.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 21.2	(k) -8.5	(l) 16.3	(m) -7.2	(n) 3.0	(o) 100	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 27.8	(d) N/A	(e) 26.1	(f) N/A	(g) 24.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.0	(b) 11.2	(c) 9.4	(e) -26.0	(f) 29.9	(g) 4.5	(h) 1.2	(i) -29.2	(j) 30.8	(k) -31.9	(l) 31.6	(m) -34.4	(n) 32.3	(o) -37.7	(p) 33.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-9.1	-8.2	-3.2	4.4	10.7	15.8	19.0	18.6	14.8	8.6	1.9	-4.2		
	DBStd	10.89	6.46	5.65	5.98	4.16	4.28	3.50	3.13	2.66	3.71	4.13	4.77	5.31		
	HDD10.0	2502	592	511	412	174	42	2	0	0	4	77	246	442		
	HDD18.3	4682	851	744	669	418	239	88	30	28	118	302	494	700		
	CDD10.0	978	0	0	1	6	64	178	279	268	147	34	2	0		
	CDD18.3	117	0	0	0	0	3	13	51	38	11	1	0	0		
	CDH23.3	616	0	0	0	0	38	109	279	151	36	3	0	0		
	CDH26.7	92	0	0	0	0	7	18	47	18	3	0	0	0		
(14) Wind	WSAvg	3.9	4.2	4.2	4.2	4.4	4.3	3.6	3.3	3.4	3.1	3.8	4.0	4.4		
(15) Precipitation	PrecAvg	863	58	40	51	62	68	74	74	86	95	94	88	72		
	PrecMax	1240	115	95	126	140	188	187	184	241	211	194	196	159		
	PrecMin	569	13	5	4	22	16	9	14	26	13	4	22	15		
	PrecStd	143	24	21	27	26	34	36	36	38	45	43	32	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	5.0	15.5	19.7	27.6	29.1	30.3	28.8	26.8	22.9	16.4	7.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	3.5	2.3	10.5	16.3	23.3	25.9	28.0	26.8	24.1	19.7	11.6	5.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	2.1	1.0	7.5	13.4	20.3	24.3	26.4	25.2	22.3	16.8	9.6	3.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	0.5	-0.2	4.4	10.9	17.7	22.3	24.9	23.6	20.7	14.6	8.0	2.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		7.7	7.8	8.7	8.1	9.6	9.4	8.6	7.5	7.7	5.9	5.5	5.9		
	5% DB	MCDBR	6.0	7.2	10.0	12.9	13.8	12.9	11.2	10.0	9.5	9.1	7.4	6.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.290	0.320	0.357	0.370	0.388	0.390	0.379	0.365	0.331	0.302	0.287		
	taud		2.333	2.288	2.322	2.327	2.372	2.344	2.358	2.385	2.422	2.503	2.505	2.359		
	Ebn at Noon		817	882	905	899	895	877	871	866	848	831	794	762		
	Edh at Noon		74	95	108	118	117	121	118	111	98	78	64	65		
(44) All-Sky Solar Radiation	RadAvg		1.09	1.99	3.27	4.46	5.55	5.98	6.00	5.07	3.80	2.22	1.17	0.85		
	RadStd		0.12	0.22	0.37	0.50	0.46	0.41	0.34	0.34	0.29	0.25	0.15	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page